Multi-rate Co-simulation Framework with Taylor-series-based Variable-step Solver for Grid-connected Power Converters*

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Abstract: Grid-connected converters (GPC) are playing an increasingly important role in distribution networks. Performing electromagnetic transient (EMT) simulations on power electronics and distribution networks can significantly improve the analysis accuracy. However, the existing simulation software struggles to handle distribution networks with a large number of power electronic switches, leading to unacceptable simulation times. To address this issue, a system-hierarchical multi-rate co-simulation framework is proposed. The system is hierarchically divided into different rate subsystems based on timescales, and solvers with different simulation rates are used to solve them separately. A Taylor-series-based variable-step solver is proposed for power electronic systems, and numerical compensation algorithms are designed for multi-rate interfaces to improve the system stability and accuracy. Compared with commercial software, the proposed framework increased the simulation speed by more than 200 times in the studied case, involving 576 switching devices and 14 bus distribution networks, while contributing less than 1% to the relative error.

Keywords: Grid-connected power converter, distribution power grid, multi-rate, electromagnetic transient, co-simulation

1 Introduction

Grid-connected power converters (GPCs) are pivotal equipment in medium-voltage distribution grids [1-2]. Capable of connecting energy from various distributed sources and loads, such as renewable generation plants and energy storage facilities, GPCs facilitate the integration of such energy into the power grid [3-4]. However, the complexity of the topology, the large number of switches, and high switching frequency in GPCs result in rapid changes in the circuit matrix [5]. Additionally, when simulating the integration of GPCs with external grid systems, the overall system combining the GPC and the distribution power grid has a typical multi-timescale hybrid characteristic [6-7]. The simulation of medium-voltage distribution networks containing numerous power electronic devices using electromagnetic transient (EMT) simulations has always been challenging. Commercial software requires a long computational time, which severely limits the research and development of GPCs.

Researchers have explored two technical approaches for improving the simulation efficiency of distribution grid systems containing complex power electronic devices. One approach is to use a multi-rate simulation architecture. The overall system simulation acceleration was achieved by dividing the system and employing different simulation rates for subsystems with different time constants. Another approach is to increase the simulation speed of the individual solvers by optimizing the performance of the model and solver, thereby improving the efficiency of solving the overall system.

The multi-rate method is widely used in large-scale system simulations [8-9]. An EMT co-simulation was introduced for large-scale AC and multi-terminal DC systems in Refs. [4, 10]. In this approach, time-varying Thevenin and Norton equivalents were proposed to build a multi-rate interface. Although this study was insightful, the proposed multirate interface was optimized for long-distance transmission-line coupling models in large-scale transmission networks, which are not present in the application scenario of the

Manuscript received October 22, 2024; revised January 27, 2025; accepted February 8, 2025. Date of publication March 31, 2025; date of current version February 17, 2025.
* Corresponding Author, E-mail: zhaozm@tsinghua.edu.cn
* Supported by the National Key Research and Development Program (2023YFB3307000).
Digital Object Identifier: 10.23919/CJEE.2025.000111

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present study. Zhan et al. [11] proposed a multi-rate, multi-core method to accelerate the simulation of large-scale power systems. This method achieves faster simulation speeds but may lead to accuracy loss in the simulation cores and rate interfaces. Additionally, this method is designed for large-scale power grid systems, and it is difficult to analyze the power electronic dynamic progress on a microsecond timescale. The interfaces of the above multi-rate methods are suitable for the dynamic processes of large-scale power grids with time scales ranging from seconds to milliseconds. In summary, multi-rate co-simulation frameworks have proven to be effective in simulating large-scale power grids [12]. However, there remains a lack of relevant research and application cases for the distribution grid system targeted in the present study, which contains numerous power electronic switches. While existing multi-rate frameworks can offer some insights for this study, they are not directly applicable to the system under investigation owing to differences in research scenarios.

Optimizing solvers to improve the simulation efficiency of power electronic systems is another effective approach. For time-varying system equations introduced by the two-state power electronic switch model, solvers using implicit integration algorithms exhibit higher stability [13]. However, owing to the influence of the two-state switch model, the implicit integration process of the solver involves repetitive matrix decomposition and inversion operations, which significantly increase the computational cost [14]. To enhance computational efficiency, latency-based methods have been introduced to decouple power electronic switches from external system parts, enabling a combination of explicit and implicit integration methods [15-16]. However, the accuracy of the latency-based method depends on the stability of the decoupling interface variables, particularly when the overall system has multi-timescale hybrid characteristics [17]. In contrast, explicit integration algorithms are less affected by time-varying system equations, providing faster overall solution speeds suitable for large-scale power electronic systems [18]. However, explicit integration requires an appropriate step size to ensure the solution accuracy and stability [14]. In summary, compared with implicit integration methods, explicit integration methods offer faster solution speeds but require careful step size selection in the co-simulation. The Taylor-series-based variable-step (TSVS) solver has been shown to be effective in solving matrix systems containing a large number of power electronic switches [19], but has not yet been applied to multi-rate co-simulations because of the complexity of step size adjustment among different solvers. When constructing a multi-rate co-simulation framework, the framework should be able to flexibly adjust the step size of the simulation algorithm to meet the precision and timing requirements of the multi-rate simulation, making it necessary to design an efficient explicit solver and provide step size adjustment methods to facilitate the construction of multi-rate simulations for power electronic systems.

The simulated GPC system, coupled with a medium-voltage distribution grid, exhibited several characteristics. The time scale of dynamic processes is in the millisecond range, and the dynamic processes of the distribution network and power electronics are difficult to decouple on a time scale. Therefore, a complete EMT simulation is required to capture the desired dynamic processes. There are numerous components, particularly power electronic switches. To achieve the EMT simulation of the GPC and distribution network, a multi-rate simulation framework must be constructed, and the solver must be optimized to accommodate the multi-rate simulation scenario. Therefore, this paper proposes a system-hierarchical multi-rate co-simulation framework that integrates the benefits of a multi-rate framework and an explicit integration solver to improve the simulation efficiency. The main features of the proposed system-hierarchical multi-rate co-simulation framework are as follows.

(1) The overall system is hierarchically layered into several subsystems according to the system time constants, and a multi-rate multi-simulator framework is used to simulate a complex system composed of the GPC and distribution grid. The proposed framework utilizes the advantages of the multi-rate technique to accelerate the simulation speed of the overall system.

(2) A TSVS solver is proposed based on the characteristics of a power electronic system. This solver adapts the step size based on the overall system

state and variable derivatives of the power electronic subsystem, which is compatible with the proposed multi-rate framework. The proposed TSVS improves the simulation efficiency of the GPC subsystems.

(3) According to the characteristics of the interface between the GPC and the distribution network, a compensation algorithm for multi-rate interface variables is specifically designed to improve the stability of the interface variable conversion between slow and fast systems, thereby reducing the multi-rate error.

The remainder of this paper is organized as follows. Section 2 proposes the hierarchical multi-rate simulation framework. Section 3 elaborates on the TSVS. Section 4 describes the multi-rate simulation system interface. Section 5 discusses the implementation of the proposed framework. Section 6 presents the simulation results and detailed analysis. Finally, Section 7 concludes the study.

2 System-hierarchical multi-rate simulation framework

A hierarchical simulation approach can effectively model complex systems on multiple timescales [20]. The proposed system-hierarchical simulation framework partitions the overall system into several subsystems, considering the timescale differences between power electronic switches and the distribution power grid. The time constant of each component of the simulated system is obtained based on its physical characteristic [20]. Subsequently, based on the simulation objectives and time constants, the system is partitioned into several subsystems in a hierarchical manner. Subsequently, the framework utilizes multi-rate techniques for different subsystems to enhance the simulation efficiency. The system features several aspects: ① Power electronic components and their control systems are placed within the fast-rate subsystem, employing the proposed TSVS solver for EMT simulation; ② The distribution power grid and its auxiliary systems are categorized within the slow-rate subsystem by utilizing a fixed-step integration algorithm for the EMT simulation; ③ Multiple simulation integration rates are applied to fast-rate and slow-rate subsystems. Coupling between the subsystems is achieved using the multi-rate interface technique introduced in Section 4.

2.1 System partitioning

System partitioning is based on the circuit substitution theorem [21]. The interface was constructed based on circuit characteristics to ensure consistency in the current or voltage constraints before and after system partitioning. During partitioning, the decoupling point is selected for the components associated with the system state variables (such as capacitor voltages or inductor currents). These components represent slow-changing aspects within a larger system, thereby enhancing the stability of multi-rate simulations [22]. Moreover, partitioning the system at the state variables maximizes the advantages of the proposed TSVS solver described in Section 3, ensuring the stability and precision of the system solutions during multi-rate simulations. In the GPC system discussed in this paper, there are numerous high-capacity parallel capacitors and series inductors distributed in multi-level cascaded structures along an AC bus or within multi-module series-parallel configurations. These characteristics establish the prerequisite circuit structure for system partitioning. Taking parallel capacitor partitioning as an example, the diagram is shown in Fig. 1.

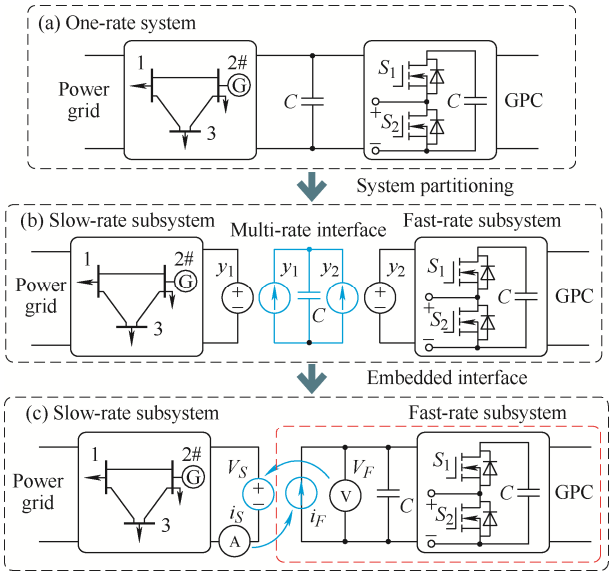


Fig. 1 System partitioning based on state variable

Fig. 1a shows the circuit diagram of the single-rate simulation system. The left side shows the power grid, the right side shows the GPC connected to grid nodes, and the middle part shows the DC output capacitor of the GPC system. As shown in Fig. 1b, following the principle of the circuit equivalent theorem, the circuit

can be partitioned into slow-rate and fast-rate subsystems using a capacitor as a boundary, interconnected through the proposed multi-rate interfaces. Controlled voltage and current sources are used to connect the two subsystems electrically. As shown in Fig. 1c, as the DC output capacitor is an integral part of the GPC system, it can be integrated with the system by embedding a multi-rate interface into the fast subsystem. Furthermore, the inductors and capacitors act as dual components of the circuits. Hence, the circuit partitioning method illustrated in Fig. 1 is equally applicable to inductor components.

2.2 System modeling

A power electronics system and power grid can be described using a state-space equation. The state-space employs state equations to model the circuits, as shown in Eq. (1), where the column vector $\boldsymbol{x}$ represents all the state variables of the system, typically comprising the voltage of the capacitor and the current of the inductor. The column vector $\dot{\boldsymbol{x}}$ denotes the derivatives of the state variables. The column vector $\boldsymbol{y}$ signifies the output variables within the circuit, encompassing all observable quantities, such as the voltages between the arbitrary circuit nodes and the currents passing through them. The column vector $\boldsymbol{u}$ represents the input variables of the system, which are typically independent power sources. $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, and $\boldsymbol{D}$ are coefficient matrices determined by the circuit topology, which can be generated using the graph-theory-based method in Refs. [22-23].

$$\begin{cases} \dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u} \\ \boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u} \end{cases} \tag{1}$$

Component models in power grids and power electronics, such as transmission lines, motors, and switches, have undergone extensive iterative development [21-22, 24]. These models can be represented using the state-space equations shown in Eq. (1).

2.2.1 Considering system-hierarchical

According to the proposed hierarchical partitioning method, Eq. (1) divides the system into two multi-rate subsystems, as shown in Eq. (2), where the subscript "s" denotes the slow-rate system and the subscript "f" represents the fast-rate subsystem.

$$\begin{cases} \dot{\boldsymbol{x}}_s = \boldsymbol{A}_s\boldsymbol{x}_s + \boldsymbol{B}_s\boldsymbol{u}_s \\ \boldsymbol{y}_s = \boldsymbol{C}_s\boldsymbol{x}_s + \boldsymbol{D}_s\boldsymbol{u}_s \end{cases} \quad \begin{cases} \dot{\boldsymbol{x}}_f = \boldsymbol{A}_f\boldsymbol{x}_f + \boldsymbol{B}_f\boldsymbol{u}_f \\ \boldsymbol{y}_f = \boldsymbol{C}_f\boldsymbol{x}_f + \boldsymbol{D}_f\boldsymbol{u}_f \end{cases} \tag{2}$$

When considering the interface between the fast- and slow-rate subsystems, the internal sources within each subsystem can be omitted. The subsystem input vector $\boldsymbol{u}$ is decided by the output vector $\boldsymbol{y}$ of other subsystems as: $\boldsymbol{u}_s = \boldsymbol{y}_f$, $\boldsymbol{u}_f = \boldsymbol{y}_s$. Then, Eq. (2) can be combined as Eq. (3).

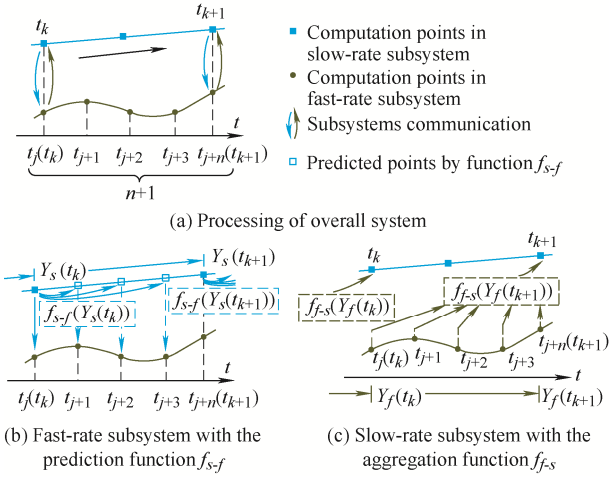
$$\begin{cases} \dot{\boldsymbol{x}}_s = \boldsymbol{A}_s\boldsymbol{x}_s + \boldsymbol{B}_s\boldsymbol{y}_f \\ \dot{\boldsymbol{x}}_f = \boldsymbol{A}_f\boldsymbol{x}_f + \boldsymbol{B}_f\boldsymbol{y}_s \end{cases} \tag{3}$$

2.2.2 Considering system multi-rate

Multi-rate simulations require strict adherence to time series constraints. As shown in Fig. 2a, the two subsystems communicate the input/output at times t_k and t_{k+1}, and they have an inner rate between t_k and t_{k+1}. When the subsystems are solved at different rates, the relationship between the inputs $\boldsymbol{u}$ of one subsystem and the outputs $\boldsymbol{y}$ of another subsystem will not be equal. We assume that the system is simulated from time t_k to t_{k+1}. which can be represented by Eq. (4).

$$\begin{cases} u_s(t_{k+1}) = f_{f\text{-}s}(Y_f(t_k)) \\ u_f(t_{k+1}) = f_{s\text{-}f}(Y_s(t_k)) \end{cases} \tag{4}$$

where $f_{f\text{-}s}$ denotes the compensation function from the fast-rate subsystem to the slow-rate subsystem, and $f_{s\text{-}f}$ represents the extrapolation function from the slow-rate subsystem to the fast-rate subsystem. As shown in Fig. 2b, the computation of the fast-rate subsystem requires extrapolation points from the slow-rate subsystem. As shown in Fig. 2c, the slow-rate subsystem requires the compensation results of the fast-rate subsystem using $f_{f\text{-}s}$. Multi-rate variable compensation is discussed in detail in Section 4.2.



(a) Processing of overall system

(b) Fast-rate subsystem with the prediction function $f_{s\text{-}f}$

(c) Slow-rate subsystem with the aggregation function $f_{f\text{-}s}$

Fig. 2 Simulation processing of multi-rate framework

2.2.3 Considering subsystem communication

The communication sequence between the subsystems determines the variables accessible to each subsystem at time t_k. In Eq. (4), $Y_f(t_{k+1})$ and $Y_s(t_{k+1})$ denote the variables collected in t_{k+1} steps from the fast-rate and slow-rate subsystems, respectively.

In serial communication, the fast-rate subsystem computes one more step than the slow-rate subsystem, making the output of the fast-rate subsystem n more computational points than that of the slow-rate subsystem, where n is the simulation point of the fast-rate subsystem between t_k and t_{k+1} as shown in Fig. 2b. The serial communication equation is as follows

$$\begin{cases} Y_f(t_{k+1}) \in \left\{ y_f(t_0), \cdots, \underbrace{y_f(t_k), \cdots, y_f(t_{k+1})}_{n+1} \right\} \\ Y_s(t_{k+1}) \in \{ y_s(t_0), \cdots, y_s(t_{k+1}) \} \end{cases} \tag{5}$$

In parallel communication, the equation will be

$$\begin{cases} Y_f(t_k) \in \{ y_f(t_0), \cdots, y_f(t_k) \} \\ Y_s(t_k) \in \{ y_s(t_0), \cdots, y_s(t_k) \} \end{cases} \tag{6}$$

where $\{y_f(t_0),\cdots,y_f(t_k)\}$ represent the simulation points of the fast-rate subsystem from start t_0 to t_k, and $\{y_s(t_0),\cdots,y_s(t_k)\}$ represent the points of the slow-rate subsystem. Combining Eqs. (1)-(6), the state-space equation of the proposed system-hierarchical multi-rate framework can be obtained as follows

$$\begin{cases} \dot{x}_s(t_k) = \boldsymbol{A}_s x_s(t_k) + \boldsymbol{B}_s f_{f\text{-}s}(Y_f(t_k)) \\ \dot{x}_f(t_k) = \boldsymbol{A}_f x_f(t_k) + \boldsymbol{B}_f f_{s\text{-}f}(Y_s(t_k)) \end{cases} \tag{7}$$

Note that the communication sequence does not include an iterative process between the two subsystems, and the multi-rate simulation can be applied with parallel computation using a serial communication sequence.

3 Taylor-series-based variable-step solver

A TSVS solver, which leverages the recursive properties of the state equations to enhance the simulation efficiency, was designed for the GPC fast-rate subsystem. Moreover, the proposed solver is adaptable to multi-rate simulations. The proposed TSVS is a variable-step variable-order solver that can dynamically adjust the simulation rate of the GPC to satisfy the accuracy and temporal logic requirements of multi-rate simulations.

A power electronic system is a typical hybrid system characterized by dynamic changes encompassing swift transient phases induced by power electronic switch actions and slower transitional phases following these actions. In the case of GPC systems with hundreds of switch components and switch frequencies ranging between 20 kHz and 50 kHz, fixed-step integration algorithms require a small step size to match the rapid changes in the transient phases, which significantly affects the simulation time of GPC systems. The proposed TSVS solver dynamically adjusts the step size (i.e., the simulation rate) based on the system states to enhance the simulation efficiency of GPC systems. Each integration routine comprised three steps: derivative computation, rate and order adjustment, and integration calculation.

3.1 Derivative computation

The state equations for simulating a GPC system (Eq. (1)) is represented in an iterative format for integration as follows

$$\begin{cases} x_f^{(o+1)}(t) = \boldsymbol{A}_f x_f^{(o)}(t) + \boldsymbol{B}_f u_f^{(o)}(t) \\ y_f^{(o)}(t) = \boldsymbol{C}_f x_f^{(o)}(t) + \boldsymbol{D}_f u_f^{(o)}(t) \end{cases} \tag{8}$$

where $x_f^{(o+1)}(t)$ represents the (o+1)-th derivative of the state variable x_f at time t of the fast-rate subsystem, and $u_f^{(o)}(t)$ and $y_f^{(o)}(t)$ are the o-th derivatives of the input variable u and output variable y at time t of the fast-rate subsystem, respectively.

According to Eq. (8), the derivatives of the state variables x within the GPC fast-rate subsystem can be computed. In proposed system-hierarchical multi-rate simulation framework, the system input variable u typically comprises two parts: the independent power sources within the GPC subsystem and the interface variables between subsystems. The independent power sources in the GPC subsystem are usually explicitly defined, such as utility three-phase power and direct current sources, which can be represented as time-varying functions $u(t)$. Therefore, high-order derivatives of $u(t)$ can be easily obtained. The interface variables between subsystems, as shown in Eq. (4), denote the input variable u of the fast-rate subsystem as the output variable y of the slow-rate subsystem. When the internal information and solving process of the slow-rate subsystem are known, the high-order

derivatives of the input variable u for the fast-rate subsystem can be directly obtained through the integration solving process of the slow-rate system, completing accurate computations for the fast-rate subsystem without loss of precision.

However, owing to constraints in the simulation software, acquiring the solution process of the slow-rate subsystem from external sources becomes challenging, making it difficult to directly obtain the high-order derivatives of the slow-rate system as inputs for the fast-rate subsystem. In such cases, during the computation of the fast-rate subsystem, the values received from the slow-rate subsystem via the multi-rate interface are considered as 0-th order derivative values, and the high-order derivative values of the input variable are all set to 0. The error of the multi-rate interface was jointly controlled using the simulation rate adjustment and interface compensation algorithms.

3.2 Simulation rate adjustment

Based on Eq. (8), the Taylor series can be expressed as

$$x_f(t_{j+1}) = x_f(t_j) + \sum_{i=1}^{p} x_f^i(t_j)\frac{(t_{j+1}-t_j)^i}{i!} + R_i(t_j) \quad (9)$$

where $x_f(t_{j+1})$ is the integration result of time step t_{j+1}, and $x_f^i(t_{j+1})$ is the i-th order derivative of the state variable x of time step t_j. p is the truncated order of the Taylor series. $R_n(t_j)$ is the remainder term of the Taylor series, which is the sum of the terms of an order higher than the truncation order p.

The simulation rate of the subsystem is determined using two factors. The first factor is the approximation error of the Taylor series, which constrains the local simulation step size. The second factor pertains to the time-sequence requirements of the multi-rate interface.

With respect to the approximation error of the Taylor series, the remaining term $R_n(t_j)$ in Eq. (9) was expanded to analyze the approximation error of the Taylor series. The approximation error, $\varepsilon_f(t_j)$, was obtained using Eq. (10).

$$\varepsilon(t_k) = x_f^{(i+1)}(t_k)\frac{(t_{k+1}-t_k)^{(i+1)}}{(i+1)!} + O((t_{k+1}-t_k)^{(i+2)}) \approx x_f^{(i+1)}(t_k)\frac{(t_{k+1}-t_k)^{(i+1)}}{(i+1)!} \quad (10)$$

where, $O((t_{j+1}-t_j)^{(i+2)})$ is the Peano remainder term. As shown in Eq. (10), the error of the state variables is obtained iteratively, which can be calculated during the integration process without additional processing.

Setting the absolute error tolerance as ε_{tol}, the optimal order p and step size ($\Delta t_\varepsilon(t_j)=t_{j+1}-t_j$) for integration should follow Eq. (11). To prevent excessive algorithmic complexity, an upper limit of the truncation order, $p<p_{\max}$, is specified. The stability of the solver in non-stiff systems can be ensured using Eq. (11).

$$\begin{cases} x_f^p(t_j)\dfrac{\Delta t_\varepsilon^p(t_j)}{p!} \leqslant \varepsilon_{tol} \\ p < p_{\max} \end{cases} \quad (11)$$

Regarding the second factor, it is essential to synchronize the local simulation step with the communication timestamps of the multi-rate interface. The proposed TSVS solver actively adjusts the order of the Taylor series and the local step size based on the communication timestamps set by the multi-rate interface to ensure alignment between the local simulation step size and the multi-rate interface step size.

In Fig. 3, $\Delta t_\delta(t_j)$ represents the time difference between the current simulation time t_j and the subsequent communication timestamp t_{k+1}. In Fig. 3a, when $\Delta t_\varepsilon(t_j)$ is smaller than $\Delta t_\delta(t_j)$, $\Delta t_\varepsilon(t_j)$ satisfies the error limitation, restricting the simulation rate owing to the local truncation error. Conversely, in Fig. 3b, $\Delta t_\varepsilon(t_j)$ exceeds $\Delta t_\delta(t_j)$. Prioritizing the multi-rate sequence alignment, the local simulation rate was determined by the communication timestamp. Overall, the proposed TSVS method adjusts the simulation rate and integration order in response to the states of the system variables.

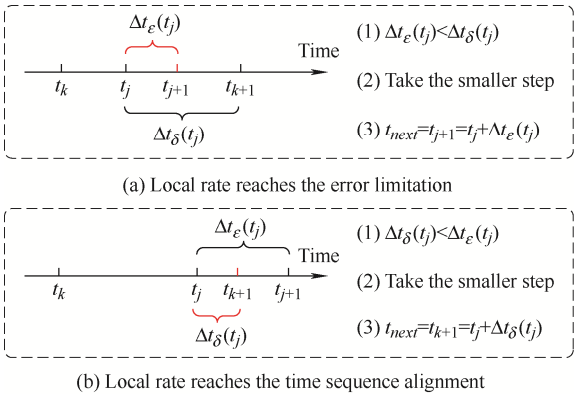


Fig. 3 Local simulation rate adjustment

3.3 Integration calculation

The solution process for the TSVS solver is illustrated in Fig 4. Note that, in the GPC system, $\boldsymbol{x}_f$ typically represents a vector of state variables with multiple elements, expressed as $\boldsymbol{x}_f = [x_{f,1}, x_{f,2},\cdots, x_{f,n}]^{\mathrm{T}}$ During the integration process, the solver operates algorithms on each element within the vector, optimizing the integration steps and the Taylor expansion order utilized. This ensures that every state variable satisfies both the precision requirements and temporal constraints of the multi-rate system.

Algorithm 1: Taylor-series based variable-step solver (TSVS)

Input: State variable $x_f(t_j)$ of local time t_j, system time t_k, subsequent system time t_{k+1}, absolute error tolerance ε_{tol}, order limitation $p_{\max}$
Output : State variable $x_f(t_{j+1})$ of local time t_{j+1}
Step 1: Obtain the current time step: $t_j \in [t_k, t_{k+1})$
Step 2: Estimate subsequent time step: t_{j+1}
Step 3: Initialize the integration order $o \leftarrow 1$
While $o \leqslant p_{\max}$ do
  Step 4: Derivative computation in Eq. (8)
  Step 5: Increment computation in Eq. (10)
  Step 6: Truncation error ε computation in Eq. (11)
  If $\varepsilon < \varepsilon_{tol}$ then
    Step 7: Obtain the $\Delta t_\varepsilon(t_j)$ and the $\Delta t_\delta(t_j) = |t_{k+1} - t_j|$
    If $\Delta t_\varepsilon(t_j) < \Delta t_\delta(t_j)$ // shown in Fig. 3 then
      Step 8: Local step $t_{j+1} \leftarrow t_j + \Delta t_\varepsilon(t_j)$
    Else
      Step 8: Local step $t_{j+1} \leftarrow t_j + \Delta t_\delta(t_j)$
    End
    Step 9: Integrate to $x_f(t_{j+1})$ by Eq. (9)
    Break;
  Else
    $o \leftarrow o + 1$;
    If $o \geqslant q_{\max}$ then reduce local step-size, and $o \leftarrow 1$;
  End
End

Fig. 4 Algorithm 1: Taylor-series based variable-step solver (TSVS)

The proposed TSVS solver consolidates the computation of step-size adjustments and the integration process within the solver, thereby eliminating the need for redundant computational resources during iterative solving. Term $\varepsilon_f(t_j)$ in Eq. (10) serves a dual purpose: it aids in the computation of increments in Step 5 of Algorithm 1 and contributes to the truncation error calculation in Step 7. Furthermore, by determining the local step t_{j+1}, the previously computed term $\varepsilon_f(t_j)$ at each order can be derived and applied to Eq. (9) to complete the integration process.

The proposed TSVS solver demonstrates advantages in solving the GPC system. In GPC systems, switch actions induce rapid changes in the system states, necessitating smaller step sizes for accurate simulation. At these points, reducing the order of the algorithm can diminish computational resource usage. Between switch actions, the system states evolve relatively slowly, permitting the use of larger step sizes to reduce the number of computational points. Simultaneously, increasing the order of the algorithm ensures the precision requirements. The proposed integrator does not exponentially increase the computational effort at higher orders. Therefore, the TSVS accurately resolves the dynamic processes in GPC systems with fewer computational resources and increased speed. TSVS is also applicable to other systems that encompass a multitude of power electronic switches.

4 Multi-rate simulation system interface

The proposed multi-rate simulation framework relies on a multi-rate system interface to accomplish the synchronization and compensation of the interface variables.

4.1 Multi-rate system synchronization

In the proposed framework, a fixed simulation step of the slow-rate subsystem (power grid) was used as the system synchronization rate, as shown in Fig. 2. The fast-rate subsystem employed the TSVS algorithm to simulate the GPC equipment. The TSVS adjusts the simulation rate based on the variable states of the subsystems. Additionally, it communicates with the multi-rate synchronization interface at synchronization time points to obtain the interface variables from the slow-rate subsystem. Simultaneously, the interface variable information of the fast subsystem is updated. Note that the proposed framework is extensible and can be applied to simulate multiple subsystems.

4.2 Multi-rate system synchronization

4.2.1 Variable prediction from slow-rate to fast-rate subsystem

As depicted in Fig. 2b, the fast-rate subsystem simulates from time step t_j to t_{j+n}, covering n computational points. However, the communication of interface variables only occurs once at communication time point t_k. To minimize the communication delay error of multi-rate subsystems, the values of the slow-rate subsystem interface variables from time t_{j+1} to t_{j+n} should be predicted.

The interface variables of the slow-rate subsystem

are extrapolated using two methods. If the solver of the slow-rate subsystem can provide derivative information, the accuracy can be enhanced by leveraging the derivative information of the interface variables. The state-space solver provides at least first-order derivative information, as shown in Eq. (1), which can be used to make a linear prediction. If the higher-order derivative information can be obtained from the slow-rate subsystem Eq. (9) can be modified into an extrapolated formula (Eq. (12)), thereby improving the accuracy of the variable prediction. The prediction accuracy was evaluated using the remainder term, as described in Eq. (10).

$$y_s(t_{j+1}) = y_s(t_j) + \sum_{i=1}^{p} y_s^i(t_j)\frac{(t_{j+1}-t_j)^i}{i!} \tag{12}$$

If the solver used in the slow-rate subsystem cannot provide derivative information for the interface variables, a linear prediction is applied using the previous simulation points. Variable prediction is expressed as follows

$$y_s(t_{j+1}) = y_s(t_j) + \frac{t_{j+1}-t_{j-1}}{t_j - t_{j-1}}(y_s(t_j) - y_s(t_{j-1})) \tag{13}$$

where $y_s(t_{j+1})$ is the extrapolated value at time point t_{j+1}.

4.2.2 Variable aggregation from fast-rate to slow-rate subsystem

As shown in Fig. 2c, when the system is simulated from time steps t_k to t_{k+1}, the fast-rate subsystem undergoes n calculation points from time steps t_j to t_{j+n}. If synchronization is based solely on the state variable $x_f(t_{j+n})$ at the last simulation point, the high-frequency switch actions inherent in the GPC will lead to outliers in the state variables transmitted from the fast-rate to slow-rate subsystem. In this study, a comprehensive algorithm for interface variables was employed to mitigate the effect of the structural changes induced by switching actions near the synchronization point on the stability of the multi-rate interface.

In Fig. 2c, the variable aggregation function $f_{f\text{-}s}$ is proposed to extract valuable information from $Y_f(t_{k+1})$. The average and weighted moving average (WMA) methods were used to build the aggregation function in a large-scale multi-rate power system simulation [17]. However, the averaging method and WMA are suitable for time series with equidistant intervals, making it difficult to handle unequally spaced time series obtained using the proposed TSVS solver. To improve this, a time-normalized exponential weighted moving average (TE-WMA) method is proposed based on the WMA as the variable aggregation function $f_{f\text{-}s}$ for the co-simulation interface. The variable aggregation result $f_{f\text{-}s}(Y_f(t_{k+1}))$ of time step t_{k+1} is given by Eq. (14).

$$f_{f\text{-}s}(Y_f(t_{k+1})) = \frac{\alpha_{j+n}\cdot y_f(t_{j+n}) + \alpha_{j+n-1}\cdot y_f(t_{j+n-1})}{\alpha_{j+n}+\alpha_{j+n-1}+\cdots+\alpha_j} + \cdots + \frac{\alpha_j \cdot y_f(t_j)}{\alpha_{j+n}+\alpha_{j+n-1}+\cdots+\alpha_j} \tag{14}$$

where α_{j+n} is the weight factor of time step t_{j+n}, $Y_f(t_{j+n})$ is the simulation value of time point t_{j+n}, and Y_f is defined in Eq. (5) and Eq. (6).

The weight factor α determines the attenuation among time sequences. The principle of the weight factor is that the time steps of the simulation points close to the co-simulation synchronization time have a larger weight than those farther away. The weight factor α_i of time step i is given by Eq. (15), where λ is the exponential amplification factor, and T_i is the time-normalized factor.

$$\begin{cases} \alpha_i = \lambda^{T_i} \\ T_i = \dfrac{t_i - t_j}{t_{j+n} - t_j} \end{cases} \tag{15}$$

The exponential amplification factor λ determines the degree of attenuation among the time sequences. λ is a pre-defined parameter that can be set based on the characteristics of the interface variables. The default value of λ was set to 10. A larger λ indicates that the aggregation value depends more on the nearest simulation point $Y_f(t_{j+n})$. The value of T_i ranges from 0 to 1, representing the normalized temporal correlation between time t_i and synchronization time t_{k+1}.

5 Framework implementation

The implementation of the proposed TSVS-based hierarchical multi-rate co-simulation framework is illustrated in Fig. 5. The overall system was divided into two subsystems according to the system time constant and co-simulation using two different solvers. The power grid was modeled and simulated using a

commercial EMT solver under a slow-rate simulation. The GPC was modeled and simulated using the proposed TSVS solver under a fast rate simulation. The multi-rate compensation component includes a fast-rate to slow-rate aggregation function $f_{f\text{-}s}$ and a slow-rate to fast-rate prediction function $f_{s\text{-}f}$. These two compensation components transfer the interface variables between two different simulation rates and maintain the stability and accuracy of the overall simulation system.

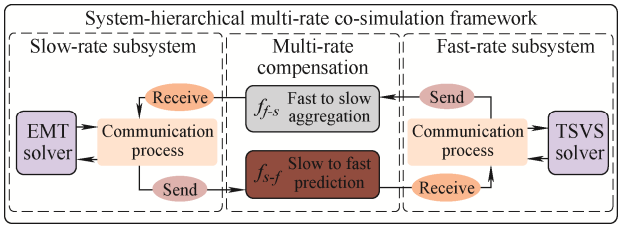


Fig. 5 Schematic of the proposed system-hierarchical multi-rate co-simulation framework

Fig. 6 illustrates the overall procedure of the proposed system-hierarchical multi-rate co-simulation framework, where the multi-rate compensation and TSVS solver for the GPC subsystem are proposed.

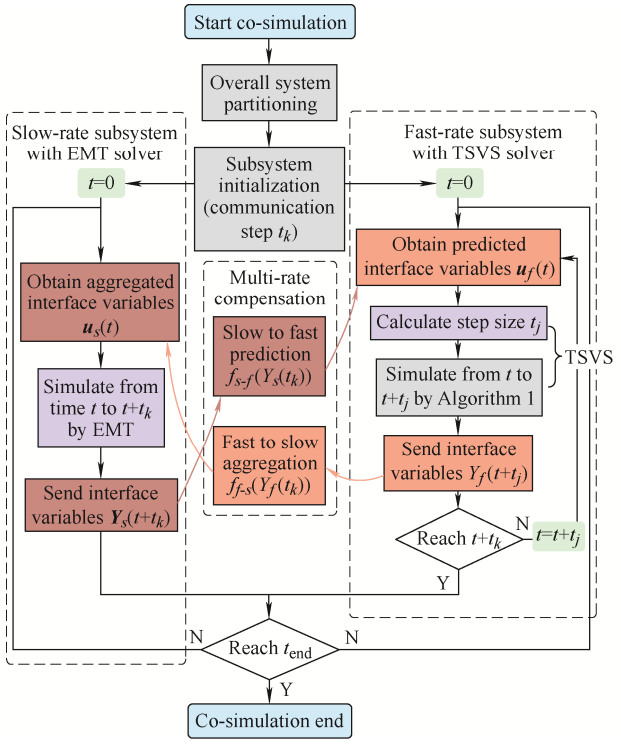


Fig. 6 Procedure of the proposed system-hierarchical multi-rate co-simulation framework

The EMT solver used for the slow-rate subsystem is a fixed solver in Matlab/Simulink. The proposed TSVS solver and multi-rate co-simulation framework were coded as a user-defined dynamic link library (DLL) using C++ and implemented in Simulink using an application programming interface (API). An ideal transformer model was used as the physical circuit connection interface [25]. The interface variables were communicated through a shared memory technique to guarantee the efficiency of the co-simulation. Note that, in the proposed framework, elaborate simulation results of the GPC can be obtained with a significantly reduced simulation time cost, which is infeasible for traditional methods [7]. A detailed comparison is presented in Section 6.

6 Case study

In this section, the proposed system-hierarchical multi-rate co-simulation framework is applied to simulate the operation of a 10 kV, 2 MW, four-port GPC within a 14-bus distribution power grid. This case verifies the accuracy of the proposed TSVS solver, the stability of the proposed multi-rate interface, and the efficiency of the overall co-simulation framework.

6.1 Case setting

The topology of the studied case is shown in Fig. 7. The 10 kV, 2 MW, four-port GPC was connected to the IEEE 14-bus system at bus 7 through the proposed multi-rate co-simulation interface. The interface variables are the voltage of bus 7 of the 14-bus power grid subsystem and the inductance current of the HVAC side of the GPC subsystem. The IEEE 14-bus system contains 5 generators, 10 local loads, and 20 lines, and the parameters of the 14-bus system were obtained from Ref. [26].

The aimed GPC features four ports: a 10 kV high-voltage AC port, a 10 kV high-voltage DC port, a 380 V three-phase low-voltage AC port, and a ±375 V low-voltage DC port. The four ports were linked through a 20 kHz high-frequency AC busbar. The detailed components and parameters of the 10 kV GPC are listed in Tab. 1. The scale of the system was extremely large, which presented significant challenges for the simulator. This case simulates the

transient electromagnetic dynamic process of the interaction between the GPC and the distribution grid over a period of 0.2 s. As shown in Fig. 7, the loads in the case study were gradually connected to the GPC. At 0 s, the 1/6 MW LVAC load was connected to the GPC. At 0.075 s, the HVDC 250 kW load was connected, and at 0.15 s, the LVDC 200 kW load was connected.

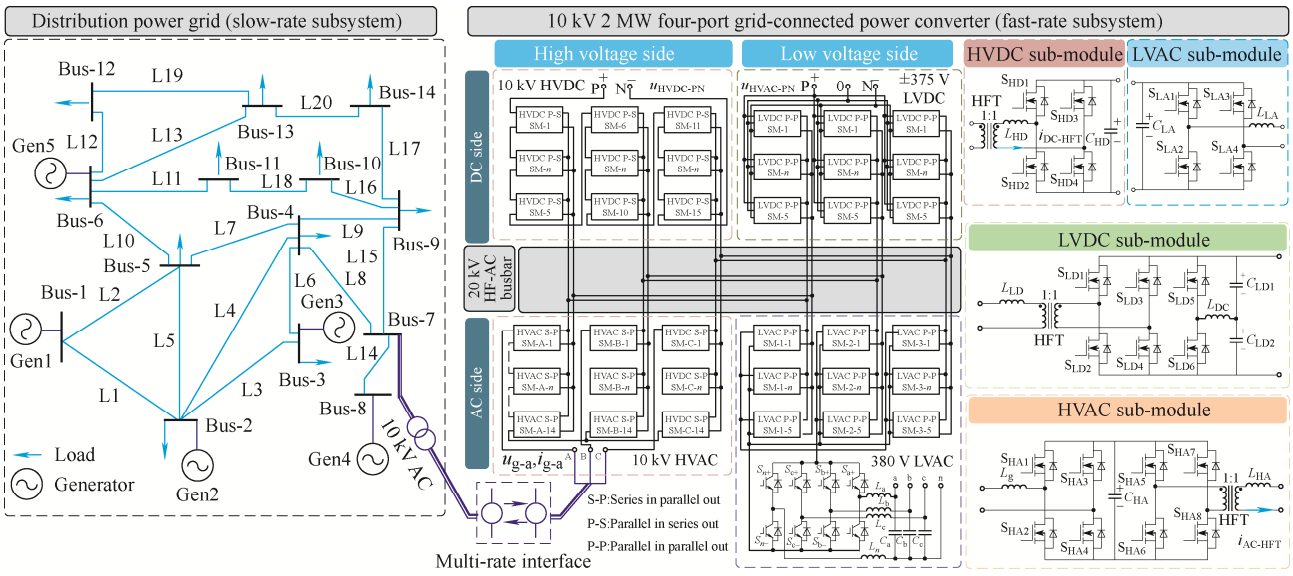


Fig. 7 Topology of the studied case: IEEE 14-bus distribution power grid with 10 kV, 2 MW, four-port GPC

Tab. 1 Schematic of the proposed system-hierarchical multi-rate co-simulation framework

Categories	Item	Quantity	Parameter
Sub-module	HVDC side	15	—
	LVDC side	15	—
	HVAC side	42	—
	LVAC side	15	—
	Total	87	—
Switching device	SiC MOSFET	544	—
	IGBT	32	—
	Total	576	—
HFT	72		Up to 20 kHz
Resistance	94		1 mΩ-400 Ω
Inductance	166		16-56 mH
Capacitance	105		250 μF-15 mF

The proposed TSVS was employed to simulate the 10 kV GPC, whereas the reference was established by the co-simulation of a commercial power electronic simulator. The detailed configuration of the case study is presented in Tab. 2. The experiments were conducted on a personal computer with an Intel Core i7-11700 processor, 64 GB of 3200 MHz RAM, running Windows 10 and Matlab 2024a.

Tab. 2 Comparisons of proposed method and commercial software

Algorithms items		Multi-rate TSVS-C[1]	Multi-rate TSVS-nC[1]	Single-rate TSVS	Ref[2]
Communication rate/s		5×10^{-4}	5×10^{-4}	1×10^{-6}	1×10^{-6}
Slow-rate subsystem	Solver	ODE4-Runge-Kutta			
	Step size/s	5×10^{-4}	5×10^{-4}	1×10^{-6}	1×10^{-6}
Fast-rate subsystem	Solver	TSVS	TSVS	TSVS	DOPRI[3]
	Max step size/s	1×10^{-6}			
	Error tolerance	1×10^{-4}			
$\text{Error}_{\text{rel}}$	$i_{\text{g-a}}$	3.37×10^{-3}	3.78×10^{-2}	8.78×10^{-4}	—
	$i_{\text{AC-HFT}}$	5.59×10^{-5}	5.58×10^{-4}	1.04×10^{-5}	
	$i_{\text{DC-HFT}}$	3.97×10^{-5}	1.93×10^{-4}	4.54×10^{-6}	
	$u_{\text{HVDC-PN}}$	4.97×10^{-4}	6.17×10^{-4}	4.21×10^{-4}	
	$u_{\text{LVDC-PN}}$	4.42×10^{-4}	5.52×10^{-4}	3.70×10^{-4}	
Simulated time/s		0.2			
Simulation time cost		115.51 s	112.91 s	348.49 s	7 h 31 min
Normalized time cost		1	0.98	3.02	234.27

[1] -C means with multi-rate compensation, -nC means without compensation.

[2] Ref is obtained by Matlab and commercial power electronic simulator.

[3] DOPRI is the non-stiff solver of commercial power electronic simulator.

6.2 Waveform comparison

To evaluate the accuracy of the proposed system-hierarchical multi-rate co-simulation framework with the TSVS solver, several critical voltage/current waveforms of different simulation settings were compared. The first waveform was simulated using the proposed TSVS with a multi-rate compensation

method, and the second waveform was obtained using the proposed TSVS without multi-rate compensation. The black reference waveform was obtained by the co-simulation using two commercial solvers. Detailed information is provided in Tab. 2. The zoomed-in waveform shows the simulation point using the three methods above, according to color.

Fig. 8 shows the waveform of the HVAC-side current $i_{g\text{-}a}$, which represents the interface variables between the two subsystems. Figs. 9 and 10 depict the current waveforms of the high-frequency transformer (HFT) on the HVAC and HVDC sides. The current waveform illustrates the microsecond-level dynamic GPC process during the load switching. The frequency of the current waveform changes very rapidly (up to 20 kHz), posing high demands on the stability of the solver for the accurate simulation of such rapid changes. Figs. 11 and 12 display the simulated load voltage waveforms from the HVDC and LVDC ports, respectively, showing the millisecond-level dynamic process. The proposed multi-rate framework maintains a high level of accuracy compared with commercial software, both in the steady and dynamic states. The waveform comparison demonstrates that the proposed

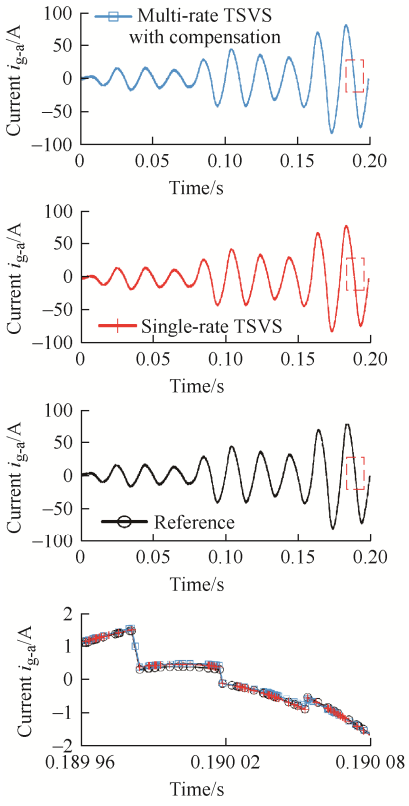


Fig. 8 Simulation result by different methods: Distribution grid-GPC interface current $i_{g\text{-}a}$ of the HVAC side

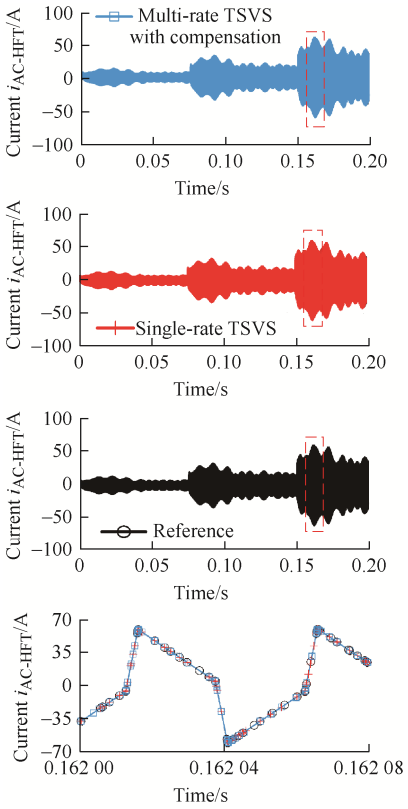


Fig. 9 Simulation result by different methods: High-frequency transformer current $i_{AC\text{-}HFT}$ of the HVAC side sub-module 1

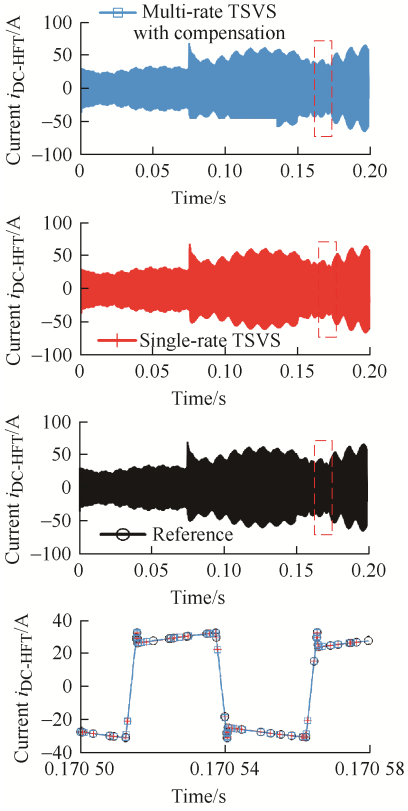


Fig. 10 Simulation result by different methods: High frequency transformer current $i_{DC\text{-}HFT}$ of the HVDC side sub-module 1

multi-rate framework, incorporating TSVS and the compensation algorithm, maintains accuracy without compromise, and is thus capable of delivering highly

precise simulation results. A quantitative comparison of the simulation accuracy and efficiency is presented in Section 6.3.

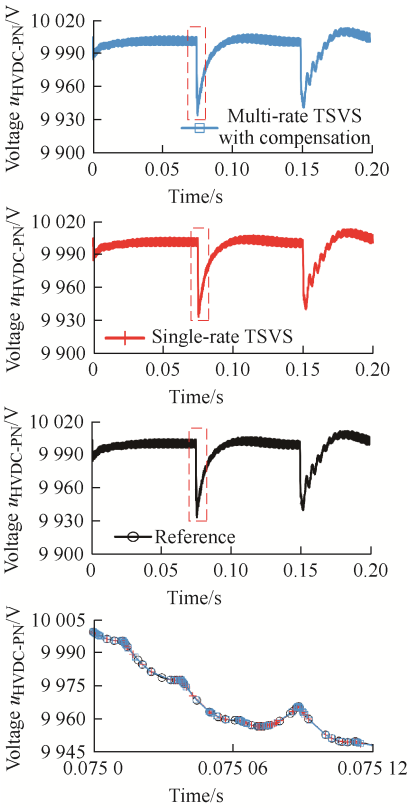


Fig. 11 Simulation result by different methods (load voltage $u_{HVDC-PN}$ of the HVDC side)

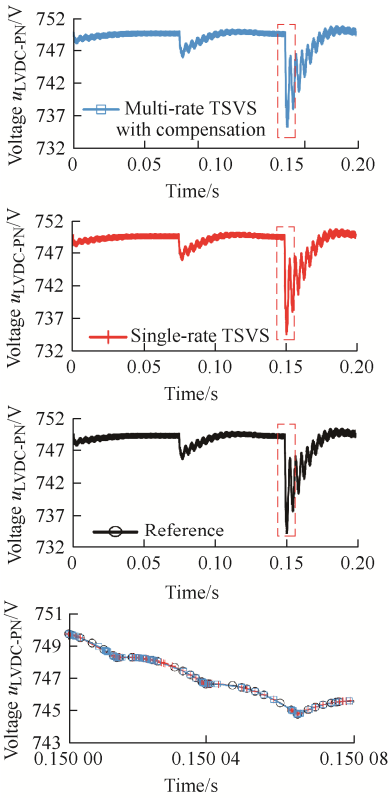


Fig. 12 Simulation result by different methods (load voltage $u_{LVDC-PN}$ of the LVDC side)

Fig. 13 shows a comparison of the multi-rate interface variable compensation algorithm proposed in Section 4. Fig. 13a shows the current waveform $i_{g\text{-}a}$ at the multi-rate interface, whereas Fig. 13b shows the voltage waveform $u_{g\text{-}a}$ at the multi-rate interface. The line in the figure shows that significant and non-smooth errors appear at the interface variable at fixed time intervals (communication step size). The interface variable waveform without deploying the multi-rate interface compensation algorithm exhibits a significant error compared with the reference waveform, which is particularly evident in the current waveform. The error directly affects the accuracy of both the slow-rate and fast-rate subsystems, leading to a decrease in the accuracy of the entire simulation system and even causing simulation divergence. The line in the figure represents the variable waveform with the proposed multi-rate interface compensation algorithm, which exhibits minimal error compared with the reference variable waveform, maintaining stability and accuracy even in high-frequency oscillations. The effectiveness of the proposed multi-rate interface compensation algorithm was further demonstrated in a zoomed-in waveform.

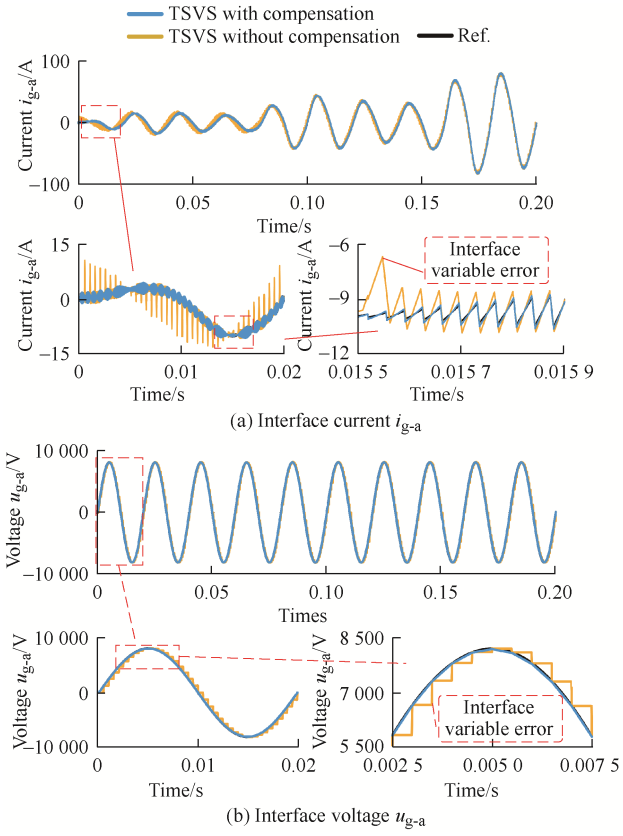


Fig. 13 Comparisons of the simulated waveform for the proposed multi-rate compensation method

6.3 Quantitative comparison

To quantitatively analyze the accuracy of the proposed multi-rate simulation framework and TSVS solver, the relative error between the simulation results of the proposed method and the reference waveform was calculated as follows

$$\text{Error}_{\text{rel}} = \frac{\left\| x - x_{\text{ref}} \right\|_2}{\left\| x_{\text{ref}} \right\|_2} \tag{16}$$

where $\text{Error}_{\text{rel}}$ is the relative error, x is the simulated waveform sequence, and x_{ref} is the reference sequence. $\| \bullet \|_2$ is the operator of Euclidean norm.

A quantitative analysis of the simulation waveforms depicted in Figs. 8-12 was performed to calculate the relative error, as detailed in the relative error section of Tab. 2. Tab. 2 indicates that the proposed TSVS solver closely aligns with the results from the commercial simulation software, with relative error magnitudes ranging from 10^{-4} to 10^{-6}. These results demonstrate that the proposed TSVS solver exhibits higher simulation accuracy and that the proposed multi-rate compensation algorithm effectively reduces simulation errors by up to one order of magnitude.

Tab. 2 presents a comparison of the simulation efficiencies. Simulating the 0.2 s dynamic process of the system using commercial simulation software would take a considerable amount of time, approximately 7 h and 31 min. In contrast, the proposed system-hierarchical multi-rate co-simulation framework, in conjunction with the TSVS solver and compensation algorithm, can reduce the simulation time to 115.51 s while ensuring accuracy, achieving a speedup of 200 times compared with commercial software. In the proposed method, the TSVS solver contributed to a speedup of 70 times and the multi-rate framework contributed to a speedup of three times.

To evaluate the performance of the TSVS solver further, Tab. 3 shows a comparison of the simulation times required by the solvers. The TSVS solver reduces the per-step computation time compared with a commercial solver. At the same level of accuracy, the simulations can be completed with a larger average step size using the proposed TSVS solver.

Tab. 3 Performance comparison between solvers of different max step settings

Solver	Max step size	Time cost/s	Point number	Time cost per point	Average step size
DOPRI	1×10^{-7}	560	109 383	0.005 120	9.14×10^{-8}
	1×10^{-6}	525	20 113	0.026 103	4.97×10^{-7}
	1×10^{-5}	520	12 914	0.040 266	7.74×10^{-7}
TSVS	1×10^{-7}	24	104 728	0.000 229	9.55×10^{-8}
	1×10^{-6}	5	15 343	0.000 326	6.52×10^{-7}
	1×10^{-5}	4	12 754	0.000 314	7.84×10^{-7}

In summary, the proposed multi-rate simulation framework can accurately and rapidly simulate complex system processes involving GPC and power distribution systems, achieving more than 200 times the simulation speed while contributing less than 1% to the simulation error in the studied case.

6.4 Applicability discussion

Note that the proposed method is not only applicable to the modeling and simulation of GPCs and distribution grids, as discussed in this paper, but also highly versatile and can be applied to similar case simulations. The framework adopts a multi-rate EMT simulation architecture. The proposed TSVS solver and compensation algorithm are optimized for systems with multiple power electronic switches. The proposed framework is particularly beneficial in areas, such as electric vehicle systems, renewable energy grids, and advanced manufacturing processes, where power electronics and control systems must be modeled together with other physical systems.

7 Conclusions

This paper presented a system-hierarchical multi-rate co-simulation framework to address the challenges in the simulation of GPCs in distributed power grids. The framework divides the overall system into several subsystems based on time scales and employs different simulation rates to accelerate the simulation. A TSVS solver compatible with the multi-rate framework was designed for the dynamically changing structure of power electronic subsystems. Compensation algorithms for multi-rate interface variables were proposed to address the difficulties in transitioning between fast and slow rates. Case studies demonstrate that, compared with the co-simulation using commercial

software, the proposed method reduces the simulation time by more than 200 times, while contributing less than 1% to the error. The proposed method can also be applied to simulate other complex systems with multiple timescales, including numerous power electronic switches.

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